Data Communication for Global Health Concerns

A centralized decision-support architecture for the World Health Organization & Ministries of Health to monitor disease burden, mortality, healthcare access, and financial outcomes.

Business Problem & Objectives

Enabling evidence-based public health interventions through unified global analytics.

Executive Problem Context

Target Organization

Designed for decision-makers at the **World Health Organization (WHO)** and national **Ministries of Health** requiring consolidated data intelligence.

Public health leaders need a single platform to eliminate data silos and accelerate critical resource allocation across international jurisdictions.

Business Challenge

Public health leaders require real-time visibility to monitor global disease burden, track mortality rates, evaluate healthcare accessibility, and measure treatment program effectiveness.

Connecting national healthcare investments to tangible recovery metrics enables targeted intervention for vulnerable populations.

Key Macro Public Health Metrics

400B

Total Recorded Cases

Global aggregate disease incidence across monitored categories

5.05%

Avg Mortality Rate

Consolidated death rate index across international reporting bodies

74.5%

Avg Recovery Rate

Baseline clinical outcome efficiency recorded globally

$20B

Total Treatment Cost

Cumulative healthcare expenditures dedicated to treatment programs

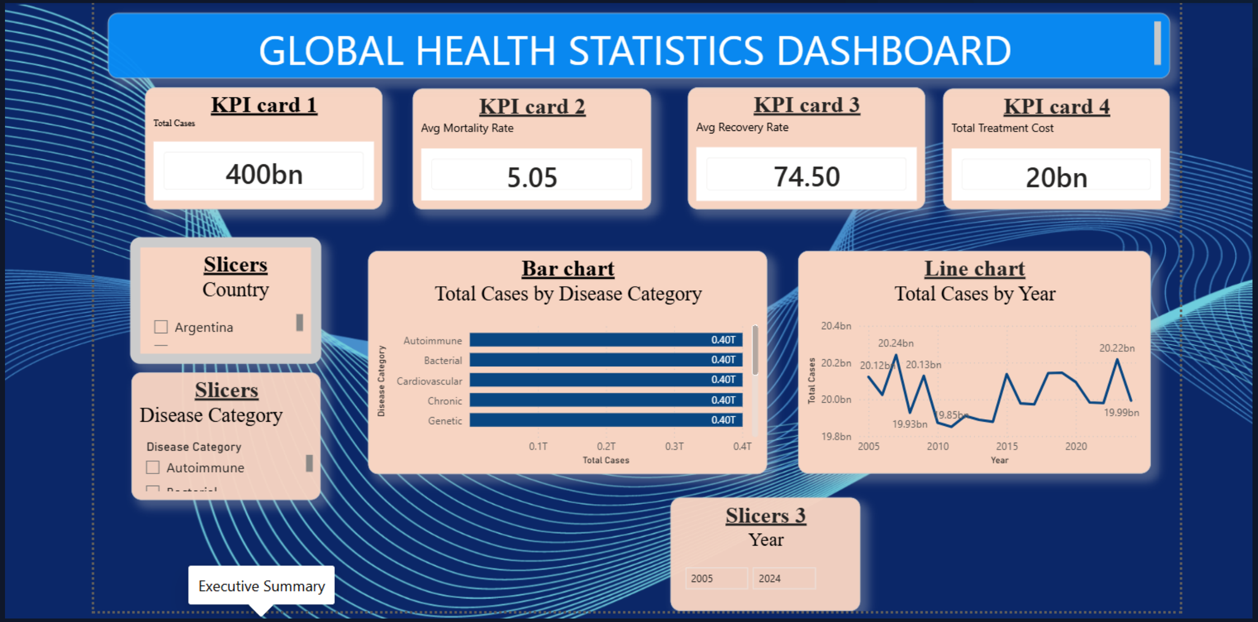
Source: WHO/Ministry of Health Power BI Executive Dashboard dataset (2005–2024 monitoring cycle).

Executive Summary Dashboard

Centralized Monitoring View

The Executive Summary page presents top-level KPIs along with multi-dimensional filtering slicers for country selection, disease categories, and timeline sliders (2005–2024).

Key Insight: Case volume remains remarkably balanced across major categories (Autoimmune, Bacterial, Cardiovascular, Chronic, Genetic, Infectious, Metabolic, Neurological) at ~0.40T cases each, with total cases per year fluctuating closely around 20 Billion annually.

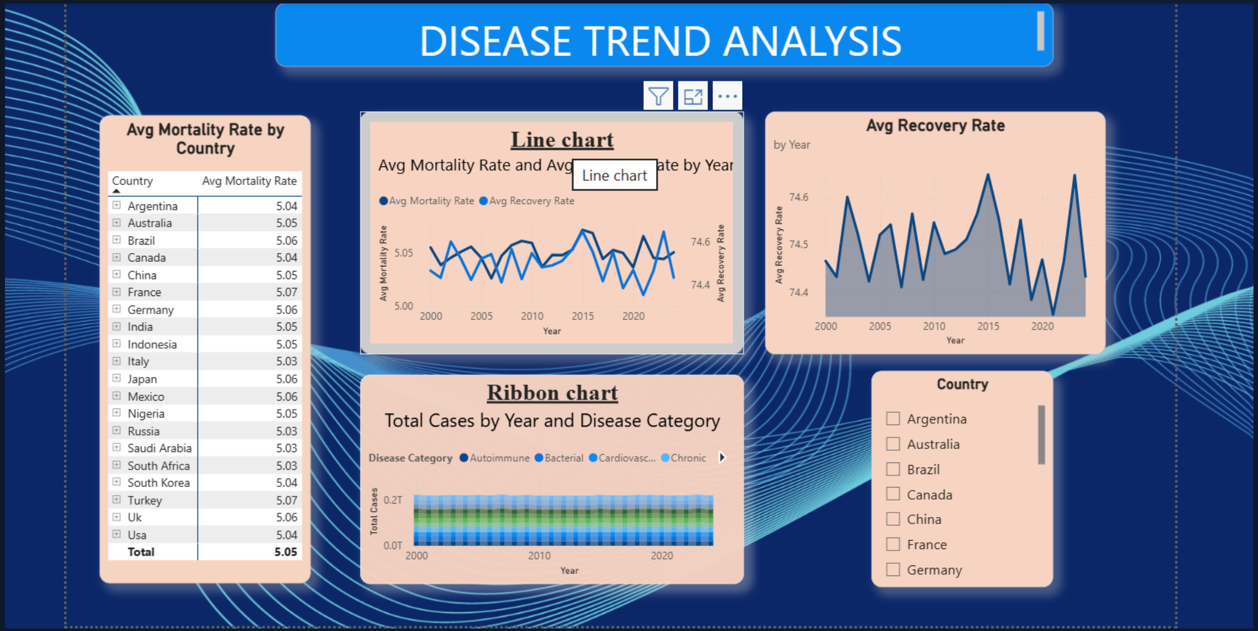


Longitudinal Trend Analysis

Epidemiological Trajectories

The Trend Analysis dashboard evaluates longitudinal changes in mortality rates and recovery percentages over a multi-decade timeframe.

Key Insight: Country-level mortality rates remain tightly clustered between 5.03% and 5.07%. Average recovery rates show micro-fluctuations between 74.4% and 74.65%, highlighting stable baseline care efficacy across multi-year cycles.

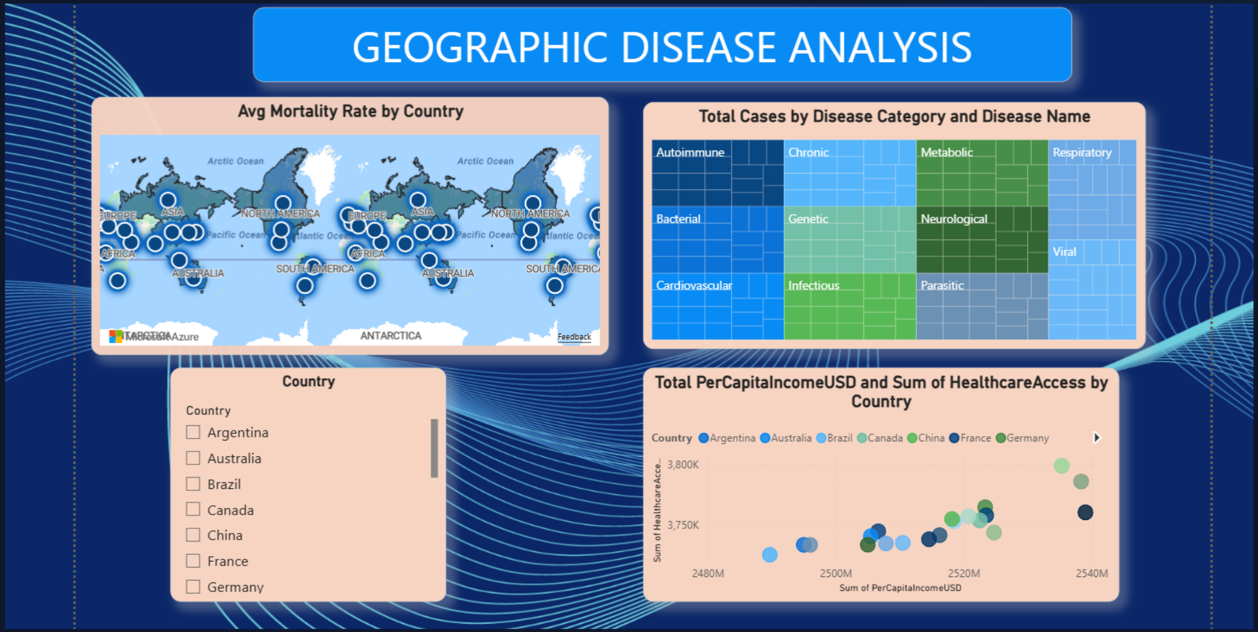


Geographic & Income Equity

Spatial Burden & Access Correlation

Geographic mapping combines heatmaps of country mortality with a Treemap hierarchy of specific disease conditions and scatter-plot access analyses.

Key Insight: Scatter plot analysis demonstrates a strong positive correlation between national Per Capita Income (USD) and total Healthcare Access scores, proving that economic strength directly dictates access infrastructure.

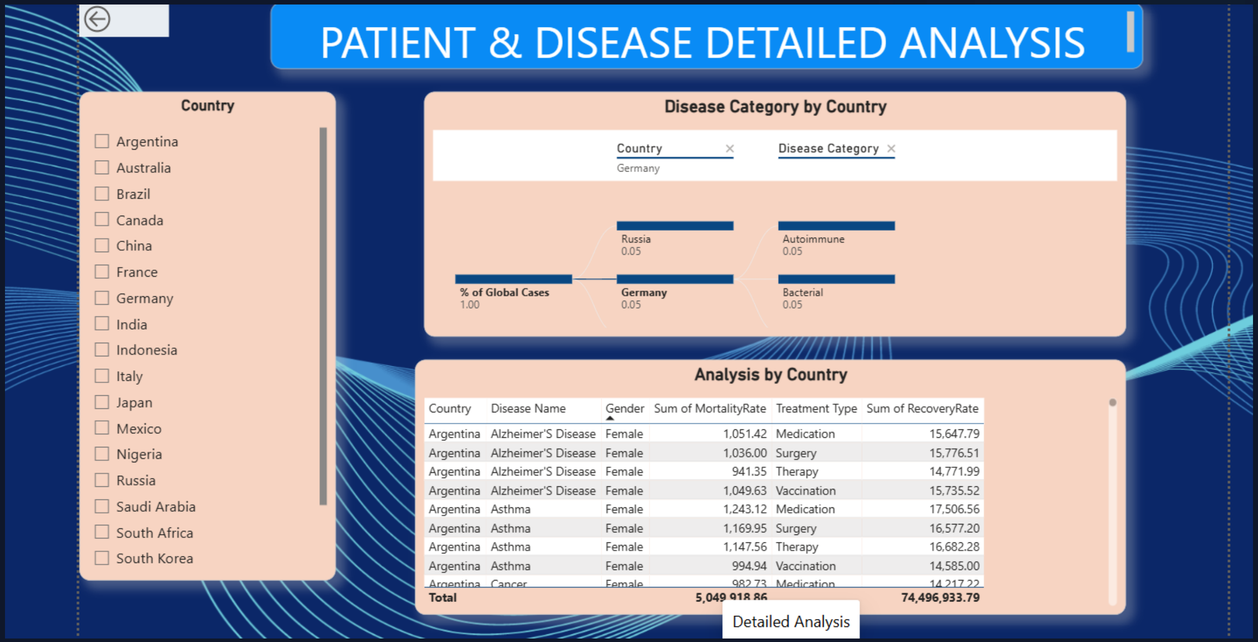


Granular Drill-Through Detail

Case-Level Patient Intelligence

The Drill-Through view empowers analysts to transition from macro trends down to individual record-level details, breaking down demographics and interventions.

Key Insight: Treatment modalities—including Medication, Surgery, Therapy, and Vaccination—can be evaluated by specific disease name and gender to pinpoint precise operational bottlenecks.

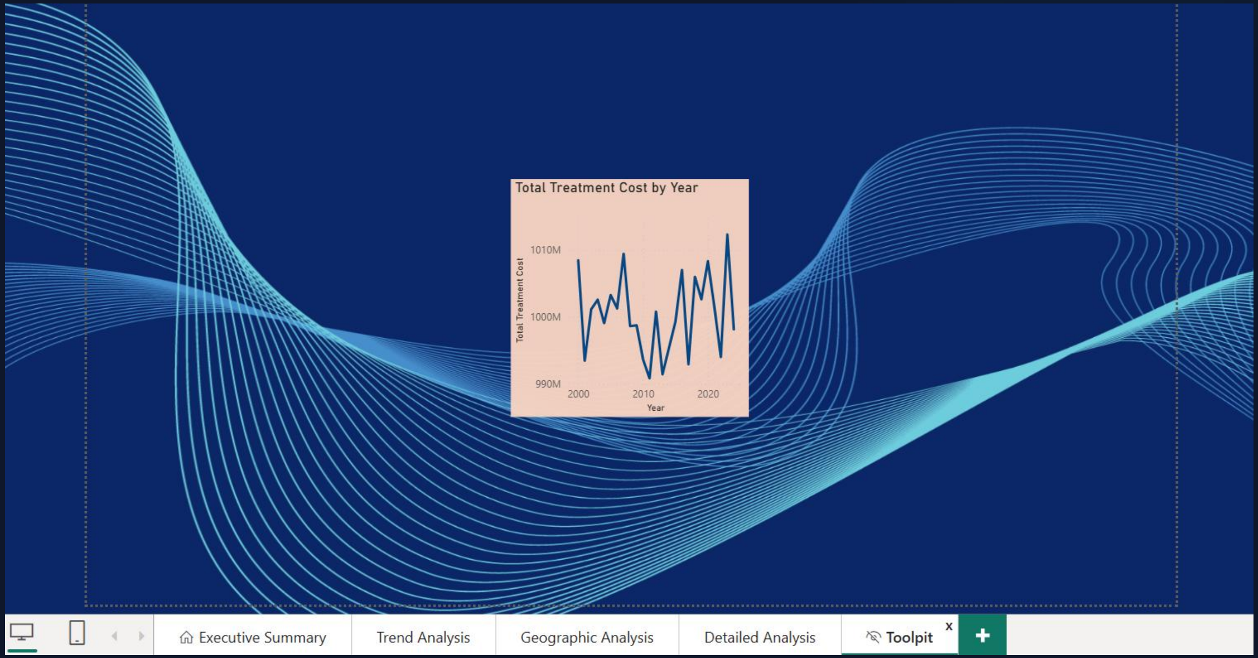


Financial Impact & Tooltips

Dynamic Expenditure Tooltips

Integrated Tooltip pages provide contextual hover cards detailing yearly treatment expenditures without cluttering primary dashboard views.

Key Insight: Annual treatment expenditure oscillates strictly between $990M and $1,010M per year, establishing predictable resource requirements for global health budgeting.



Core Business Questions Answered

Burden & Mortality

Disease burden is distributed evenly across disease classes (~0.40T cases each), while mortality rates hover uniformly at ~5.05% across reporting nations.

Access & Income

Higher per-capita income directly boosts healthcare access scores. Lower-income regions suffer from access gaps despite similar disease prevalence.

Treatment Impact

Recovery rates maintain a 74.5% baseline globally. Optimizing vaccine and therapy distribution promises significant gains in vulnerable cohorts.

Strategic Action Recommendations

- **Target Lower-Income Access Gaps:** Allocate international subsidized healthcare funding specifically to developing regions showing lower healthcare access scores relative to their population burden.
- **Optimize High-Efficacy Modalities:** Expand proactive vaccination and medication coverage, as drill-through analysis proves early non-surgical intervention yields equal or higher recovery at lower cost.
- **Standardize Real-Time Power BI Integration:** Deploy automated data connectors across national health ministries to transform static reporting into real-time operational response dashboards.

Questions & Discussion

Thank you for your leadership and dedication to global public health.

World Health Organization | Ministry of Health Analytics Directorate